

MA 301
Final
Spring - 2013

Problem	points	out of
1		10
2		10
3		10
4		10
5		10
6		14
7		15
8		18
9		13
Score		100

Name: _____

HR: _____

Signature: _____

READ BEFORE YOU START

- * Show all work.
- * Change your calculator mode to RADIANS.
- * Sloppy solutions may be penalized.
- * Use calculators ONLY for computations.
- * You must write up every problem in a clear and organized way.
- * If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Given $F = [2, 2y + z, y + 1]$

(a) Show that F is conservative in two different ways
method 1

method 2

(b) Evaluate $\int_C F \cdot dr$ when C is a straight line from $(-1, 0, 2)$ to $(0, 2, -6)$

Problem 2: Let $F = [y, x \sin y]$ and C be the boundary (with positive orientation) of the region between $y = \frac{\pi}{2}x$, $y = \frac{\pi}{2}$ and $0 \leq x \leq 1$ in the xy -plane.

Sketch C and compute $\oint_C F \cdot dr$

$$N_x = \sin y$$

$$M_y = 1$$

Apply Green's Thm :

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S (\sin y - 1) dA$$

$$= \int_0^1 \int_{\frac{\pi}{2}x}^{\frac{\pi}{2}} (\sin y - 1) dy dx$$

$$= \int_0^1 -\cos y - y \Big|_{\frac{\pi}{2}x}^{\frac{\pi}{2}} dx$$

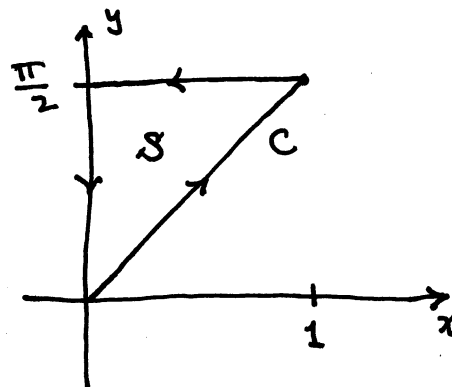
$$= \int_0^1 \left(-\cos\left(\frac{\pi}{2}\right) - \frac{\pi}{2} \right) - \left(-\cos\left(\frac{\pi}{2}x\right) - \frac{\pi}{2}x \right) dx$$

$$= \int_0^1 \cos\left(\frac{\pi}{2}x\right) + \frac{\pi}{2}x - \frac{\pi}{2} dx$$

$$= \frac{2}{\pi} \sin\left(\frac{\pi}{2}x\right) + \frac{\pi}{4}x^2 - \frac{\pi}{2}x \Big|_0^1$$

$$= \frac{2}{\pi} \sin\left(\frac{\pi}{2}\right) + \frac{\pi}{4} - \frac{\pi}{2} - 0$$

$$= \frac{2}{\pi} - \frac{\pi}{4} \approx -0.149$$



Problem 3: Let $F = [x, ye^z, e^z]$ and S be the surface (with outward orientation) of the box $|x| \leq 1, |y| \leq 2, 0 \leq z \leq 1$.

Compute $\iint_S F \cdot n \, dA$

$$\operatorname{div} \vec{F} = 1 + e^z + e^z = 1 + 2e^z$$

Apply Divergence Theorem

$$\begin{aligned} \iint_S \vec{F} \cdot \vec{n} \, dA &= \iiint_V \operatorname{div} \vec{F} \, dV \\ &= \int_{-1}^1 \int_{-2}^2 \int_0^1 (1 + 2e^z) \, dz \, dy \, dx \\ &= \int_{-1}^1 \int_{-2}^2 (z + 2e^z) \Big|_0^1 \, dy \, dx \\ &= \int_{-1}^1 \int_{-2}^2 (1 + 2e^1) - (2e^0) \, dy \, dx \\ &= \int_{-1}^1 \int_{-2}^2 (2e - 1) \, dy \, dx \\ &= (2e - 1)(4)(2) \\ &= 16e - 8 \\ &\approx 35.49 \end{aligned}$$

Problem 4: Let $f(x) = \begin{cases} 0 & -2 < x < -1 \\ 1+x & -1 < x < 0 \\ 1-x & 0 < x < 1 \\ 0 & 1 < x < 2 \end{cases}$ and $f(x) = f(x+4)$

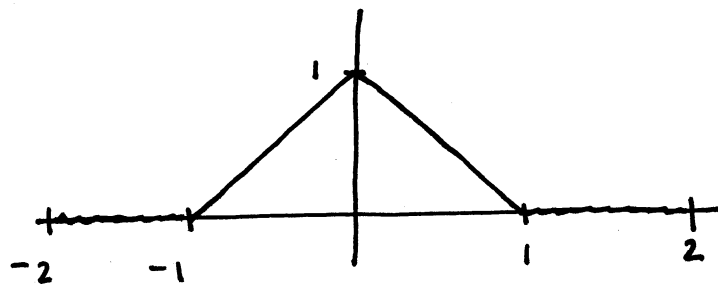
Find the Fourier Series of $f(x)$

$$a_0 = \frac{1}{2} \int_0^2 f(x) dx$$

$$= \frac{1}{2} \int_0^1 (1-x) dx$$

$$= \frac{1}{2} \left[x - \frac{1}{2} x^2 \right] \Big|_0^1$$

$$= \frac{1}{2} \left(1 - \frac{1}{2} \right) = \frac{1}{4}$$



f is even $\Rightarrow b_n = 0$

$p = 4 \Rightarrow L = 2$

$$a_n = \frac{2}{2} \int_0^2 f(x) \cos\left(\frac{n\pi x}{2}\right) dx$$

$$= \int_0^1 (1-x) \cos\left(\frac{n\pi x}{2}\right) dx$$

$$= \left[\frac{2(1-x)}{n\pi} \sin\left(\frac{n\pi x}{2}\right) + \frac{2}{n\pi} \int \sin\left(\frac{n\pi x}{2}\right) dx \right]_0^1$$

$$= \frac{2(1-x)}{n\pi} \sin\left(\frac{n\pi x}{2}\right) - \frac{4}{n^2\pi^2} \cos\left(\frac{n\pi x}{2}\right) \Big|_0^1$$

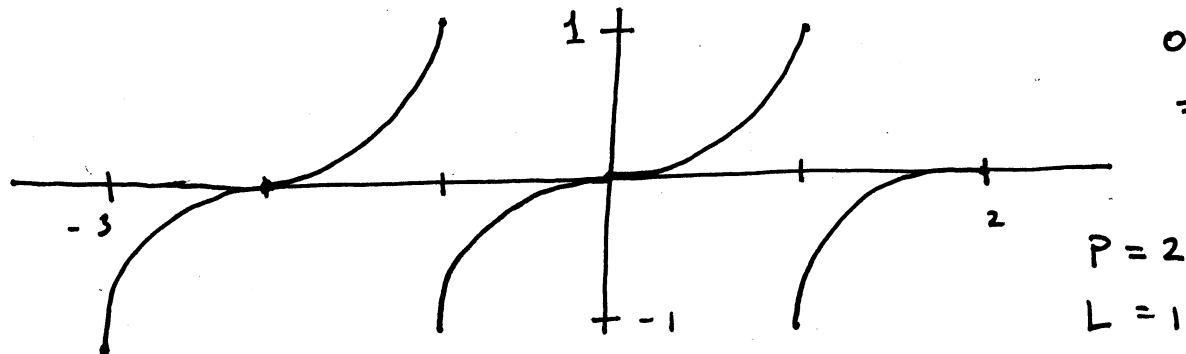
$$= 0 - \frac{4}{n^2\pi^2} \cos\left(\frac{n\pi}{2}\right) - \left(0 - \frac{4}{n^2\pi^2} \right)$$

$$= -\frac{4}{n^2\pi^2} \cos\left(\frac{n\pi}{2}\right) + \frac{4}{n^2\pi^2} = \frac{4}{n^2\pi^2} \left(1 - \cos\left(\frac{n\pi}{2}\right) \right)$$

$$f(x) = \frac{1}{4} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \left(1 - \cos\left(\frac{n\pi}{2}\right) \right) \cos\left(\frac{n\pi x}{2}\right)$$

Problem 5: Let $f(x) = \begin{cases} -x^2 & -1 < x < 0 \\ x^2 & 0 < x < 1 \end{cases}$ and $f(x) = f(x+2)$

(a) Graph $f(x)$ on the interval $[-3, 2]$



(b) Can we say that any of Fourier coefficients a_0 , a_n , or b_n are zero? Why or why not?

$$\text{ODD} \Rightarrow a_0 = a_n = 0$$

(c) **Set-up**, but **DO NOT** compute the integral(s) for all the non-zero Fourier coefficients.

("f(x)" should not be part of the answer and the integrals should be simplified as much as possible)

$$\begin{aligned} b_n &= 2 \int_0^1 f(x) \sin\left(\frac{n\pi x}{1}\right) dx \\ &= 2 \int_0^1 x^2 \sin(n\pi x) dx \end{aligned}$$

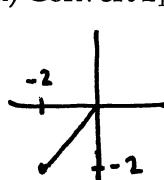
Problem 6: Let $z_1 = -2 - 2i$, $z_2 = i$. Answer the following:

(a) $z_1 z_2 = (-2 - 2i)i = -2i - 2i^2 = 2 - 2i$

(b) $z_1 - z_1 z_2 = -2 - 2i - (2 - 2i) = -4$

(c) $|z_1 z_2| = \sqrt{4 + 4} = \sqrt{8} = 2\sqrt{2}$

(d) Convert z_1 to Euler's Polar form:

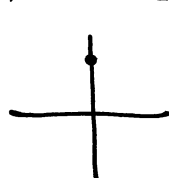


$$\theta = \frac{5\pi}{4} \text{ or } -\frac{3\pi}{4}$$

$$r = 2\sqrt{2}$$

$$\Rightarrow z_1 = 2\sqrt{2} e^{i\frac{5\pi}{4}}$$

(e) Convert z_2 to Euler's Polar form:



$$\theta = \frac{\pi}{2}$$

$$r = 1$$

$$\Rightarrow z_2 = e^{i\frac{\pi}{2}}$$

(f) (Polar Form) $(z_2)^{76} = 1^{76} e^{38\pi i}$

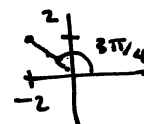
(g) (Polar Form) Solve for z : $z_2 z^{1/3} = z_1$

$$z^{1/3} = \frac{z_1}{z_2} = \frac{-2 - 2i}{i} = \frac{-2i - 2i^2}{-1} = \underbrace{-2 + 2i}_{\theta = \frac{3\pi}{4}, r = 2\sqrt{2}}$$

$$\Rightarrow z = (-2 + 2i)^3$$

$$= (2\sqrt{2} e^{i\frac{3\pi}{4}})^3$$

$$= (2\sqrt{2})^3 e^{i\frac{9\pi}{4}} \text{ or } 54\sqrt{2} e^{i\frac{9\pi}{4}}$$



Problem 7: Determine the convergence of each series. If convergent, find the sum. Justify all work.

$$(a) \sum_{n=1}^{\infty} \frac{n^2}{n-1} \quad \lim_{n \rightarrow \infty} \frac{n^2}{n-1} = \infty \neq 0$$

$\Rightarrow$ divergent

$$(b) \sum_{n=1}^{\infty} \frac{(-6)^n}{5^{n+1}} = \frac{1}{5} \sum_{n=1}^{\infty} \left(-\frac{6}{5}\right)^n \quad |r| = \frac{6}{5} > 1 \Rightarrow \text{diverges by G.S.T.}$$

$$(c) \sum_{n=1}^{\infty} \sin\left(\frac{\pi n}{2n-1}\right) \quad \lim_{n \rightarrow \infty} \sin\left(\frac{\pi n}{2n-1}\right) = \sin\left(\frac{\pi}{2}\right) = 1 \neq 0$$

$\Rightarrow$ diverges

$$(d) \sum_{n=0}^{\infty} \frac{3^{2n}}{10^n} = \sum_{n=0}^{\infty} \left(\frac{9}{10}\right)^n = \frac{1}{1 - \frac{9}{10}} = 10 \quad \text{converges by G.S.T.}$$

$$(e) \sum_{n=4}^{\infty} \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}$$

$$\begin{aligned} \sum_{n=4}^N \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} &= \left(\frac{1}{\sqrt{4}} - \frac{1}{\sqrt{5}}\right) + \left(\frac{1}{\sqrt{5}} - \frac{1}{\sqrt{6}}\right) + \left(\frac{1}{\sqrt{6}} - \frac{1}{\sqrt{7}}\right) + \cdots + \left(\frac{1}{\sqrt{N-1}} - \frac{1}{\sqrt{N}}\right) + \left(\frac{1}{\sqrt{N}} - \frac{1}{\sqrt{N+1}}\right) \\ &= \frac{1}{\sqrt{4}} - \frac{1}{\sqrt{N+1}} \end{aligned}$$

$$\begin{aligned} \text{So } \sum_{n=4}^{\infty} \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} &= \lim_{N \rightarrow \infty} \sum_{n=4}^N \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \\ &= \lim_{N \rightarrow \infty} \left(\frac{1}{\sqrt{4}} - \frac{1}{\sqrt{N+1}}\right) = \frac{1}{\sqrt{4}} \quad \text{converges} \end{aligned}$$

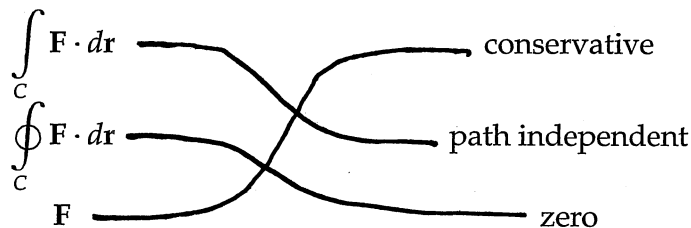
Problem 8: Answer the following true/false questions

T / ☒ F	The line integral of any vector field $\mathbf{F}$ around any closed curve C is zero
T / ☒ F	If $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for one particular closed path, the $\mathbf{F}$ is path-independent.
☒ T / F	If $\mathbf{F}$ is path-independent, then there is a potential function for $\mathbf{F}$
☒ T / F	If $\int_C \mathbf{F} \cdot d\mathbf{r} \neq 0$ for some closed path C , then $\mathbf{F}$ is path-dependent.
☒ T / F	If $\mathbf{F} = \nabla f$ then $\mathbf{F}$ is conservative.
☒ T / F	If $\mathbf{F}$ is path-independent and C is any closed curve, then $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$
T / ☒ F	If S_1 is a rectangle with area = 1 and S_2 is a rectangle with area = 2, then $2 \iint_{S_1} \mathbf{F} \cdot d\mathbf{A} = \iint_{S_2} \mathbf{F} \cdot d\mathbf{A}$
T / ☒ F	Suppose $\mathbf{F} = [1, 1, 1]$ and S is a closed surface with positive orientation. Then $\iint_S \mathbf{F} \cdot \mathbf{n} dA > 0$.
☒ T / ☒ F	Let S be a sphere of radius 1 centered at the origin and oriented outward. If $\iint_S \mathbf{F} \cdot d\mathbf{A} = 0$, then $\mathbf{F} = \mathbf{0}$
T / ☒ F	Let S be the surface of a cube centered at the origin with side length 2 and oriented outward. If $\mathbf{F} = 2\mathbf{i} - \mathbf{k}$ then the flux of $\mathbf{F}$ through S is 27.
T / ☒ F	If $\mathbf{F}$ is a vector-field such that $\text{div}\mathbf{F} = 3$ and S is a closed surface oriented outward, then $\iint_S \mathbf{F} \cdot d\mathbf{A}$ is equal to one-third the volume enclosed by S .
T / ☒ F	Suppose $f(x) = f(x + 6)$ for all x , then we say f is periodic with period 3.
T / ☒ F	$\sqrt{-5} \cdot \sqrt{-5} = 5$.
☒ T / F	If $\sum_{n=1}^{\infty} a_n$ converges, then so does $\sum_{n=100}^{\infty} a_n$.
☒ T / F	If $\sum_{n=1}^{\infty} a_n$ diverges, then so does $\sum_{n=100}^{\infty} a_n$.
☒ T / F	If $\sum_{n=10}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} \cos(a_n) = 1$.
T / ☒ F	If $\sum_{n=1}^{\infty} a_n$ converges, then so does $\sum_{n=1}^{\infty} \frac{1}{(a_n)^2}$.
☒ T / F	If $\sum_{n=1}^{\infty} a_n$ converges, then it is possible for the partial sums to converge to $\frac{3}{2}$.

Problem 9: Answer the following questions

(a) Let $\mathbf{F}$ vector field such that $\text{curl } \mathbf{F} = \mathbf{0}$.

Match each expression in the left column with the most appropriate term in the right column.



(b) Match each the Theorem in the left column with the most appropriate statement in the right column.

Divergence Theorem _____ The surface of a full sphere of radius 4

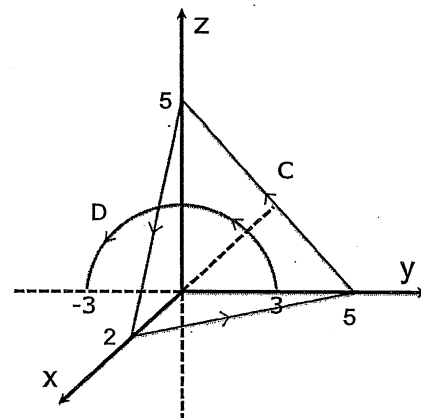
Stokes' Theorem _____ The surface of a half-sphere of radius 4

Green's Theorem _____ A disk of radius 4

(c) Let $\mathbf{F} = \nabla \left(\frac{x^2}{y+z+1} \right)$ and let C and D be shown in the figure. Compute

$$\int_C \mathbf{F} \cdot d\mathbf{r} = 0 \quad \text{since } C \text{ is closed}$$

$$\int_D \mathbf{F} \cdot d\mathbf{r} = \left. \frac{x^2}{y+z+1} \right|_{(0,3,0)}^{(0,-3,0)} = 0 - 0 = 0$$



(d) Suppose $\sum_{k=1}^{\infty} \left(\frac{c}{5} \right)^k = \frac{c}{6}$. Find all values of c which satisfy this expression.

$$-1 < \frac{c}{5} < 1$$

$$\Leftrightarrow -5 < c < 5$$

$$\frac{c}{5} \cdot \frac{1}{1 - \frac{c}{5}} = \frac{c}{5} \left(\frac{5}{5-c} \right) = \frac{c}{5-c} = \frac{c}{6}$$

$$\Leftrightarrow 6c = 5c - c^2$$

$$\Leftrightarrow c^2 + c = 0$$

$$\Leftrightarrow c(c+1) = 0$$

$$\Leftrightarrow c = 0, -1 \quad \text{both satisfy } -5 < c < 5$$